FlatHill

Robot hardware, built for scale.

A U.S. robotics hardware supply platform.
Proprietary joints. Manufacturing expertise.
An AI workflow from concept to production.

The bottleneck is repeatable hardware.

A working robot becomes a business when its cost, quality and supply can repeat across every production batch.

01	**Manufacturing expertise**	The team brings prior robot delivery experience into design decisions, supplier selection and factory processes.
02	**Owned module architecture**	Micro drives and larger joint families provide reusable building blocks across hands, arms and mobile platforms.
03	**A U.S. production path**	Selected global inputs feed documented U.S. manufacturing, supported by product-specific sourcing and authorization work.

ROSE turns each engineering and production cycle into reusable manufacturing knowledge.

An established hardware market. A focused point of entry.

NORTH AMERICAN ROBOT ORDERS / 2025

$2.25B

36,766 robots ordered

A3 reported order value. Industry context, not FlatHill's addressable component market.

$1.166B H1 2026 order value
+6.6% year over year

FLATHILL'S U.S. BEACHHEAD

Modular actuation for robot builders

Start with dexterous manipulation and joint integration programs where hardware teams need cost control, customization and a production path.

Direct buyers	Robotics OEMs and applied-AI engineering teams
Initial offers	Micro drives, joint evaluation kits and hand / arm development hardware
Expansion	Integrator programs, qualified mobile platforms and manufacturing services

A3: 2025 orders, published 6 Feb 2026 · A3: H1 2026 update, published 11 Aug 2026. Different time periods. Do not add these figures or treat them as FlatHill revenue potential.

Three buyer groups.
One component-led sales strategy.

BUYER	INITIAL SALE	EXPANSION
Robotics OEMs	Paid joint evaluation kits, integration support and a qualified production SKU.	More units of the same module, then additional joints and robot programs.
AI & research teams	Hand / arm development kits, spares, interfaces and paid setup support.	Additional rigs, replacement modules and later ROSE workflow pilots.
Automation integrators	Configured joint, hand or arm assemblies. Mobile platforms enter after product-specific qualification.	Replication across sites and service inventory, then manufacturing capacity and ROSE.

Commercial strategy and proposed customer segmentation. Named accounts, signed orders and conversion rates are not established in the supplied materials.

A team shaped by robot delivery.

150+

robots delivered in prior career programs

Team-provided track record in patrol robotics.
Not FlatHill shipments.

Hardware architecture

Integrated engineering across joint modules, compact mechanisms, drive electronics and complete robot platforms.

Production engineering

Prior robot programs span design iteration, supplier coordination, production transfer and field delivery.

Embedded systems and control

Team materials describe a decade of experience in embedded hardware, sensor fusion and motion planning.

The engineering bench spans mechanisms, motor control, embedded electronics and system integration.

Internal: original DexHand deck, team slide 3. Career experience is management-provided, not independently audited.

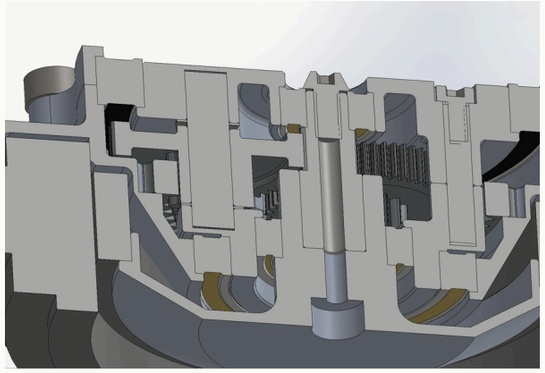
One actuation foundation. Two scales of motion.



MICRO DRIVE FAMILY

M01 / M01 Neo / M02

Compact joint architecture for fingers and small end effectors. Integrated transmission, sensing and control.



LARGE JOINT FAMILY

WZ53 / WZ79 / WZ119

Reusable joint architecture for larger mechanisms. Source design targets span 7 / 15 / 60 N·m.

Self-developed module designs allow FlatHill to control interfaces, tolerances, firmware and production changes.

Internal: joint module development PDF and DexHand deck. Large-joint figures are design targets. Production maturity varies by SKU.

Cost control starts before the factory.

A COMMON COST MODEL FOR EVERY RELEASED SKU

Landed parts + Build & test + Yield loss + Field support

Measure cost per good unit. Keep this operating-cost model separate from the origin-content calculation.

01	**Fewer unique parts**	Reuse actuator families, connectors and fixtures across products. Reduce tooling, inventory and redesign work.
02	**Design for the process**	Match tolerances to function. Standardize assembly sequence and end-of-line tests before scaling orders.
03	**Measure the delivered cost**	Track purchase cost, duties, yield loss, rework and warranty cost for each released configuration.

The economic goal: lower total delivered cost with repeatable quality.

Four platforms. Shared engineering.



01

Dexterous hands

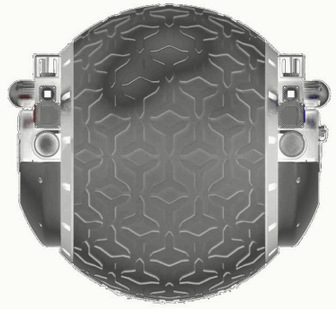
Micro drives and replaceable finger modules.



02

Quadrupeds

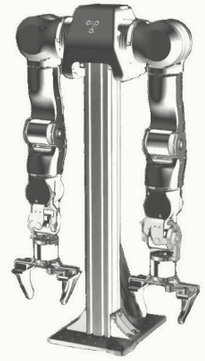
Joint assemblies for mobile inspection platforms.



03

Spherical robots

Enclosed mobility with drive and stabilization systems.



04

Dual-arm platforms

Manipulation systems with modular joints and end effectors.

Platform families demonstrate integration. Standard modules create the repeatable supply business.

Existing FlatHill product renders. Specifications, readiness and authorization are configuration-specific.

A clear path from first kit to recurring supply.

STAGE	REVENUE	ADVANCE WHEN
01 Evaluate	Evaluation-kit revenue	The buyer accepts the engineering results and funds the next integration step.
02 Design in	Engineering fees + pilot hardware	The customer approves the design, supplier qualification and production acceptance criteria.
03 Repeat	Recurring component / assembly sales	Repeat orders and contribution economics justify dedicated U.S. capacity.
04 Expand	ROSE subscriptions + manufacturing fees	Customers complete paid workflows and the cells meet quality and unit-economics targets.

Revenue scales with customer production.

Production programs	**5**
Robots per program / year	**500**
Modules per robot	**12**
Module selling price	**$180**
Hardware gross margin	**30%**

ILLUSTRATIVE ANNUAL HARDWARE REVENUE

$5.40M

5 programs × 500 robots × 12 modules × $180

Module shipments

30,000

Gross profit before operating costs

$1.62M

A steady-state production cohort, not a forecast, pipeline or TAM estimate. Prices, volumes and margins are unverified assumptions. No kit, engineering, software or whole-robot revenue is added here.

Editable illustrative scenario. Gross profit excludes operating expenses, taxes and financing costs. Assumptions require actual customer programs and a costed SKU.

U.S. manufacturing. A disciplined global input strategy.

PHASE 1 / NOW THROUGH 2027 / PLAN

01

Source selected inputs

Qualify commodity electronics and motor inputs, including eligible China-sourced parts. Record maker, origin and firmware provenance.

02

Manufacture in the U.S.

Build qualifying component assemblies and end products through U.S. operations or approved manufacturing partners.

03

Release a documented SKU

Tie supplier declarations and cost records to serial numbers, tests and the applicable authorization path.

>75% U.S.-supplier purchasing objective

80% Illustrative U.S.-qualifying component-cost target

These are separate measures. A U.S. invoice alone does not establish U.S. manufacture or component origin. Actual sourcing remains to be validated.

FAR 25.101 and 25.003. See appendices for calculation, assumptions and product scope.

Compliance is a product release discipline.

80%

Illustrative domestic component-cost target.
Supported by actual U.S. manufacture and an evidence file for each SKU.

01	**Classify the finished product**	The current FCC determination targets defined mobile robotic devices. Fixed industrial robots are excluded from that definition.
02	**Validate manufacture and component cost**	The referenced FAR test generally requires U.S. manufacture and domestic component cost above 65% through 2028, then above 75% from 2029. Exceptions require review.
03	**Complete the authorization path**	Covered List treatment and normal equipment authorization remain separate release checks. Conditional approval is a product-specific alternative where available.

FCC DA 26-786, Appendix C · FAR 25.101 · FCC DA 26-996. Regulatory snapshot: 12 Oct 2026.

Design control makes localization possible.

FlatHill’s proposed advantage is a transferable engineering package: module designs, drawings, firmware and manufacturing know-how.

01	**Own the engineering decisions**	Management reports self-developed joint module families and complete technical drawings. Confirm assignments and third-party licenses per product.
02	**Build the U.S. rights package**	Maintain an inventor and employer chain of title, U.S. filing strategy and product-specific freedom-to-operate analysis.
03	**Keep capitalization transparent**	Management states a U.S.-first financing and commercialization strategy. Confirm the FlatHill entity’s cap table separately from team members’ prior ventures.

Investors can diligence the rights, the design revisions and the manufacturing transfer together.

Management representations, October 2026. Executed IP assignments, U.S. filing receipts and a counsel FTO opinion were not reviewed for this deck.

ROSE connects engineering to manufacturing.

PHASE 2 / COMMERCIALIZATION AFTER 2027 / ROADMAP

ROSE

DRAFTGPT WORKFLOW 01

From scratch to demo

Turn requirements into a buildable design and a tested prototype.

DRAFTGPT WORKFLOW 02

From demo to mass production

Turn a working design into a controlled manufacturing package.

Management reports an early software prototype. The workflows shown here define the intended product scope, not verified shipped functionality.

From a requirement to a testable prototype.

STEP	OUTPUT	RELEASE GATE
Requirements	A requirements record with measurable acceptance tests.	An engineer approves the assumptions and test plan.
Build plan	CAD interfaces, control scaffolding and a sourcing list.	The selected modules meet the stated operating envelope.
Review	A review record with exceptions and required design changes.	Engineers resolve critical findings before authorizing a physical build.
Demo	A tested demo, measurement log and next-iteration list.	The prototype demonstrates the agreed task under defined conditions.

Interactive explanation of the planned workflow. This is not a demonstration of verified ROSE software functionality.

From a demo to a repeatable production release.

STEP	OUTPUT	RELEASE GATE
DFM	A revised design, process plan and list of critical characteristics.	Production engineering approves the process and tolerance assumptions.
Suppliers	A controlled BOM and approved supplier list.	Cost, origin and supplier qualification records support the released SKU.
Control plan	Work instructions, station tests and a traceability schema.	The pilot team can execute the same process and record the same evidence.
Pilot lot	A lot record, nonconformance actions and release recommendation.	Authorized engineers approve volume production or request another pilot.

Interactive explanation of the planned workflow. This is not a demonstration of verified ROSE software functionality.

The U.S. factory scales by proven process.

Start with flexible production. Automate the stable, repetitive work as demand and process data justify investment.

01

Manufacturing partners

Qualify machining and assembly capacity. Validate fixtures, work instructions and origin records.

Gate: repeatable pilot lots

02

Dedicated U.S. cells

Bring critical assembly, calibration and test under direct process control.

Gate: demand + yield + unit economics

03

Lights-out processes

Automate selected stable operations with monitoring, exception handling and recovery.

Gate: sustained unattended reliability

ROSE links the released design to the actual factory process and its performance data.

Every production cycle improves the next.

PHASE 3 / LONG-TERM VISION

RSI

Recursive improvement of robot hardware and its manufacturing process

ROSE uses design, test and production outcomes to propose the next controlled revision.

Observe	Yield, torque, thermal behavior, failures and field returns
Propose	Design alternatives, process settings and approved part substitutions
Validate	Simulation, bench testing and controlled pilot lots
Release	Engineer approval, versioning, traceability and rollback

Success means lower cost per good unit and faster validated changes, with safety and quality held constant.

Three stages, with measurable release gates.

STAGE	BUILD	ADVANCE WHEN
01 **Now–2027** Hardware supply	Module evaluation kits U.S. manufacturing transfer Documented sourcing	Paid pilots convert to repeat orders. Released SKUs pass cost, quality and origin review.
02 **2028–2029** ROSE + U.S. cells	Two commercial software workflows Dedicated production capacity	Customer projects complete both workflows. Cell economics support additional automation.
03 **2030 onward** Adaptive manufacturing	Broader lights-out processes Validated improvement loops	Successive released revisions improve unit cost and yield with controlled reliability.

Planning sequence. Timing depends on product readiness, demand, financing and regulatory requirements.

Capital converts engineering into U.S. supply.

WORKING SEED FINANCING PLAN

$8–10M

Existing September 2026 financing range.
Final round terms remain subject to confirmation.

Fund the first repeatable U.S. production program and validate ROSE with real engineering work.

$10M PLANNING CASE

Factory and production equipment	$2M
Team and operating runway	$3M
R&D, materials and production transfer	$3M
Working-capital / contingency reserve	$2M

Capital deployment follows pilot demand and production gates. At the lower end, stage equipment purchases and reserve use.

Next evidence: paid pilots, repeat orders, qualified component records and measured production yield.

Internal: fundraising.json, updated 5 September 2026. Allocation is a planning case, not approved spend or a funded budget.

The hardware foundation for robot builders.

Manufacturing experience today.

ROSE-enabled engineering next.

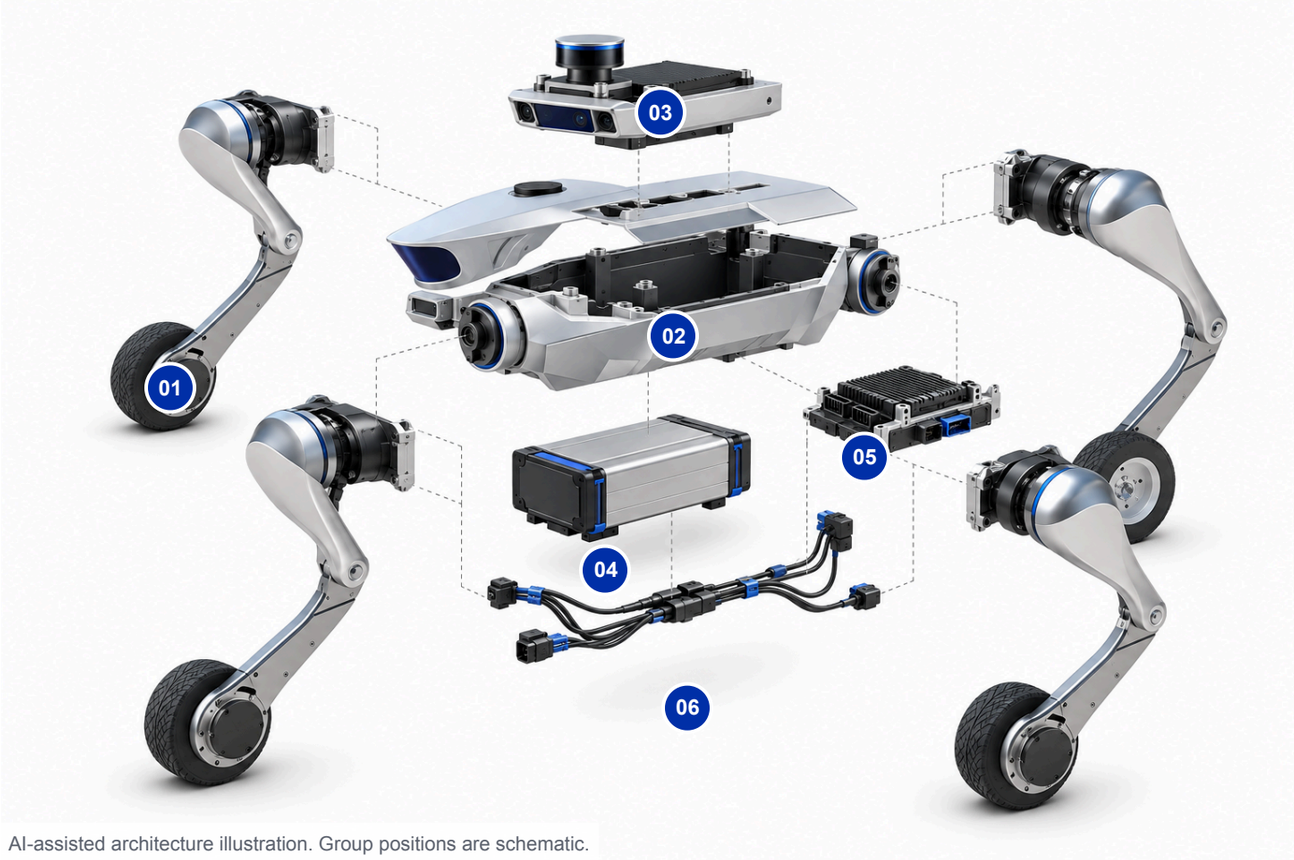
A U.S. factory that improves with every release.

FlatHill jysong47@gmail.com

U.S. hardware supply platform

Quadruped: localize the highest-value assemblies.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Joint / leg assemblies	42%	42 pp
02 Chassis and structures	16%	16 pp
03 Compute and sensing	14%	10 pp
04 Battery assembly	12%	12 pp
05 Control / power boards	10%	0 pp
06 Harness and connectors	6%	0 pp
Total	**100%**	**80 pp**

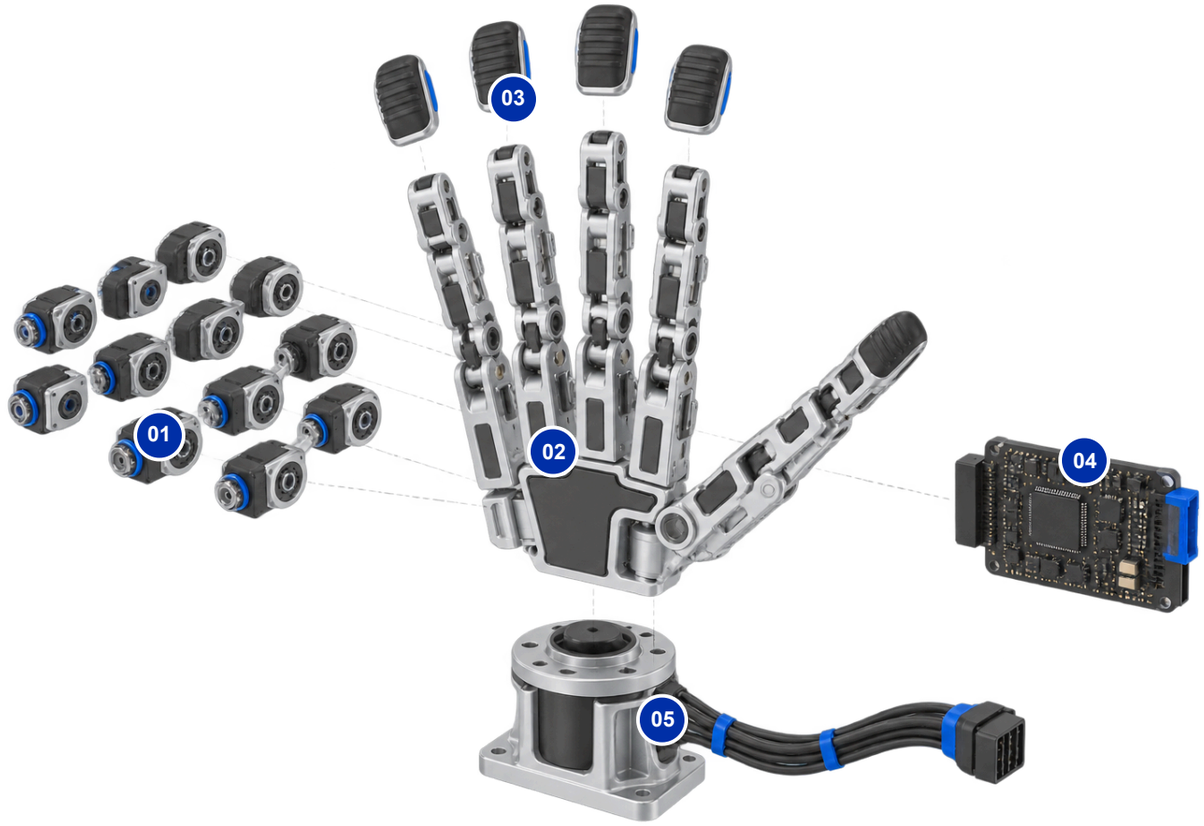
*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

Joint assemblies are the largest dependency. Imported finished joints cannot become domestic merely through a U.S. reseller.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Dexterous hand: cost lives in the micro joints.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Micro joint modules	48%	48 pp
02 Finger / palm mechanics	22%	22 pp
03 Tactile / force sensing	12%	2 pp
04 Control electronics	10%	0 pp
05 Wrist and harness	8%	8 pp
Total	**100%**	**80 pp**

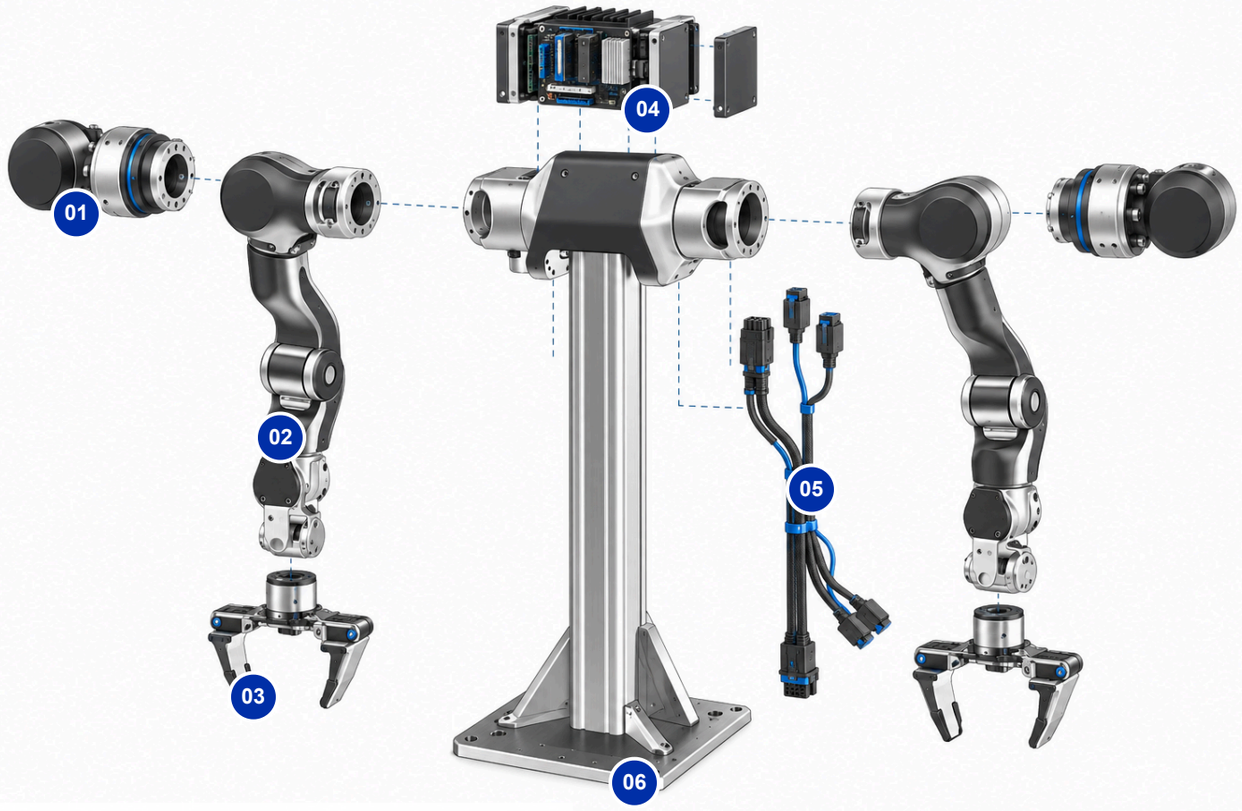
*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

A shared micro-drive design concentrates cost engineering, calibration and replacement parts in one module family.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Dual arms: reuse joints, links and test fixtures.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Joint modules	44%	44 pp
02 Links and housings	18%	18 pp
03 End effectors	12%	12 pp
04 Control / compute	10%	0 pp
05 Power / harness	10%	0 pp
06 Pedestal and base	6%	6 pp
Total	**100%**	**80 pp**

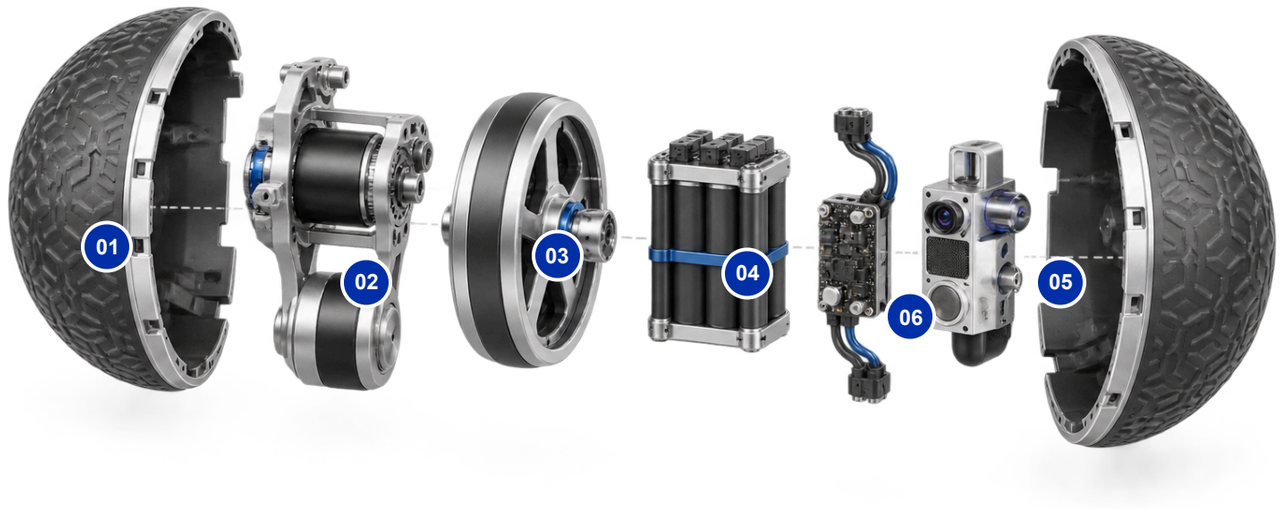
*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

A fixed industrial arm configuration has a different FCC scope analysis from a mobile robot carrying the same arm.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

Spherical robot: separate mobility from payload.

ILLUSTRATIVE DESIGN BUDGET / NOT A QUOTED OR AUDITED BOM



AI-assisted architecture illustration. Group positions are schematic.

ASSEMBLY GROUP	BOM	U.S. TARGET*
01 Shell and structure	24%	24 pp
02 Pendulum drive assembly	24%	24 pp
03 Stabilization / flywheel	14%	14 pp
04 Battery assembly	18%	18 pp
05 Compute / sensing	12%	0 pp
06 Power boards / harness	8%	0 pp
Total	**100%**	**80 pp**

*Percentage points of total component cost. Domestic contribution assumes each underlying component qualifies.

A common drive core can support multiple sensing payloads. Origin and authorization follow the shipped configuration.

Planning assumption: grouped component costs total 100%. Origin classification requires a detailed, non-overlapping SKU BOM. Final assembly cost is excluded.

The 80% model has an explicit dependency.

QUADRUPED DESIGN SCENARIO

80 / 100 = **80%**

Sum the cost of qualifying domestic direct components. Divide by the cost of all direct components.

Use one component boundary. Avoid counting both an assembly and its internal parts. Group totals in this deck are presentation rollups.

80% 5 percentage points above the 75% comparison line.

01 **Same denominator**

Exclude final robot assembly cost, sales margin and corporate R&D from the component-cost denominator.

02 **Evidence per component**

Record manufacturer, manufacturing location, cost basis and origin support. Treat unknown origin as foreign.

03 **A supplier address is insufficient**

The 42-point joint contribution is conditional. If it fails, the illustrated domestic share falls to 38%.

FAR 25.003 and 25.101. Illustrative non-iron/steel-dominant, non-exempt cost scenario. No COTS waiver assumed.

Commodity inputs are an engineering choice.

META / MUSE GADGETS

An official ESP32 example

Meta’s open-source Muse Gadget SDK supports off-the-shelf ESP32 boards and Linux devices.

It demonstrates an available development ecosystem. It does not establish FlatHill robot authorization or future component eligibility.

Official repository ↗

TESLA / GLOBAL SUPPLY

Global sourcing can coexist with U.S. manufacturing

Tesla’s 2025 annual report describes parts from thousands of global suppliers. Its 2019 filing explicitly discussed U.S. production exposure to duties on Chinese components.

These are sourcing examples, not a robotics compliance precedent or a current China-content estimate.

2025 annual report ↗

Qualify each MCU, power device and motor input against the rules and product configuration in force at release.

Meta Muse Gadget SDK · Tesla 2025 10-K · Tesla 2019 Q2 10-Q.

Evidence and assumptions stay distinct.

Source-supported materials

Internal product drawings, module concepts, team profiles and prior financing plan. Official FCC notices, FAR provisions, Meta repository and Tesla filings.

Management representations

Module IP ownership, current team participation, FlatHill financing / promotion history and the early ROSE prototype.

Planning assumptions

All four BOM allocations, the 80% domestic target, roadmap dates, manufacturing sequence and future software revenue model.

Product release documentation

Obtain supplier origin records and a costed BOM. Verify component boundaries, U.S. manufacture and applicable authorization.

IP diligence

Review assignments, third-party licenses, U.S. filing records and counsel’s product-specific FTO opinion. Patent registration alone does not establish FTO.

U.S. origin marketing

The FTC’s unqualified “Made in USA” standard requires all or virtually all U.S. content. This deck’s 80% model does not establish that claim.

Full source notes and the editable planning data accompany this presentation. Status snapshot: 12 October 2026.

USPTO: patent rights · FTC: Made in USA Rule · Full source notes.